On Optimal Modality Compression for Bandwidth Constrained Multimodal Fusion

Mengchen Fan, Fei Zhao, Chengcui Zhang, Shuya Feng, and Baocheng Geng
University of Alabama at Birmingham
Birmingham, AL, USA
Emails: fanm@uab.edu, larry5@uab.edu, czhang02@uab.edu
fengs@uab.edu, bgeng@uab.edu

***Abstract*—Multimodal fusion under communication constraints requires careful allocation of limited bandwidth across heterogeneous modalities. While it is commonly assumed that more communication resources should be allocated to higher quality modalities, this heuristic lacks rigorous theoretical justification, and it remains unclear whether it is optimal under constrained communication. In this paper, we present a theoretical analysis of modality compression strategies in bandwidth constrained multimodal fusion. By modeling compression as an additional source of uncertainty, we theoretically show that, under Gaussian assumptions, assigning compression to the modality with higher observation noise improves the overall fusion performance. We further validate this principle on both multivariate and image datasets, and observe consistent performance improvements under asymmetric compression strategies. These findings provide a principled foundation for communication aware multimodal system design and offer practical guidelines for resource constrained applications such as edge intelligence and distributed sensing.**

***Index Terms*—Multimodal fusion, modality compression, information bottleneck**

I. Introduction

Multimodal fusion [1]–[4] has become a central paradigm in modern machine learning and intelligent sensing systems, where heterogeneous sources such as images, signals, and sensor streams are jointly exploited to improve downstream decision making [5]–[7]. By combining complementary information across modalities, multimodal systems often achieve higher accuracy, greater robustness, and better reliability than their single modality counterparts [8,9]. This advantage has motivated broad interest in multimodal learning across a wide range of settings, including distributed sensing, edge intelligence, and resource-aware collaborative systems, where multiple modalities are observed at different nodes and must be fused under system constraints [10]–[12].

Despite these advantages, practical multimodal fusion is constrained. In many real-world deployments, the participating devices cannot transmit all raw observations in full due to limited bandwidth, energy, memory, or latency budgets [13]–[16]. As a result, one or more modalities must first be compressed through an encoder into some bottleneck representation or reduced dimensional message before being transmitted for fusion [17]. This introduces a fundamental trade-off between communication efficiency and information preservation. Existing studies do not explicitly address how compression should be assigned across heterogeneous modalities under a fixed communication budget. Prior work on multimodal learning has studied representation learning, cross-modal interaction, bottleneck architectures, and robustness to incomplete modalities [18]–[21]. Compression is usually treated as a generic architectural component, not as a decision variable whose assignment can fundamentally affect the final fusion performance. Therefore, a key theoretical gap remains in the literature. There is still no clear characterization of how *modality quality*, *compression induced information loss*, and *fusion performance* interact with one another when the total communication budget is fixed. In practice, compression is often assigned based on signal quality, with the higher quality modality preserved and the lower quality modality compressed [22]–[24]. While this heuristic seems reasonable, it is largely based on intuition rather than rigorous analysis, and it remains unclear whether it is always optimal under bandwidth constrained fusion [25]–[27].

This gap motivates the central question of this paper: *under a fixed communication budget, which modality should be compressed to maximize the final fusion performance?* To answer this question, we consider the two modality fusion system shown in Fig. 1. Two modalities, denoted by L and S, observe the same phenomenon of interest (PoI), that is, the same underlying source or signal to be inferred, but they may have different levels of observation quality. Due to limited communication bandwidth, only a compressed representation of one modality can be transmitted to the other modality, which acts as the fusion center. This leads to two possible communication structures. In Fig. 1(a), modality L is compressed and transmitted to modality S. In Fig. 1(b), modality S is compressed and transmitted to modality L. Although the two structures appear symmetric at the architectural level, they are fundamentally different once the modalities have unequal quality, because the impact of compression depends on which modality absorbs the compression loss.

In this work, we address this problem by developing a theoretical framework for bandwidth constrained multimodal fusion under heterogeneous modality quality. We model compression as an additional source of uncertainty and analyze its influence on the final fused decision from statistical signal processing and information theory. Under Gaussian assumptions, our analysis yields an important conclusion that *compressing* *the modality with higher observation noise leads to better final fusion performance than compressing the higher quality modality under the same communication budget.* This result provides a principled answer to the modality assignment problem and fills a gap that has not been theoretically analyzed in prior multimodal fusion literature.

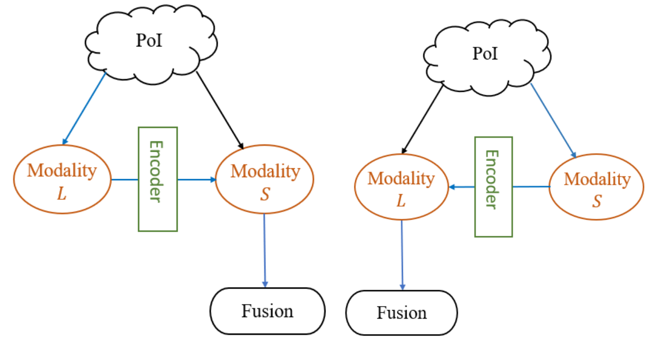
Beyond theory, an important question is whether this principle continues to hold in practical deep multimodal systems, where the encoder, communication bottleneck, and fusion model are learned jointly. To examine this, we design controlled experiments on both multivariate and image datasets. Our experimental design directly mirrors the theoretical comparison that the total communication budget is kept fixed, while only the assignment of compression is varied across modalities. This controlled setup allows us to isolate the effect of modality wise compression and test whether the theoretical principle is observable beyond the simplified Gaussian setting. The results consistently validate the theory. Together, the theoretical analysis and experimental validation establish a coherent framework for understanding communication aware multimodal fusion and provide actionable guidance for designing bandwidth constrained multimodal systems.

The main contributions of this work are summarized as follows:

- We formulate the problem of modality wise compression assignment in bandwidth constrained multimodal fusion and identify a fundamental theoretical gap in existing literature: prior work has not explicitly analyzed which modality should be compressed under heterogeneous modality quality and a fixed communication budget.
- We develop a theoretical framework that models compression as an additional source of uncertainty and show, under Gaussian assumptions, that compressing the modality with higher observation noise yields better final fusion performance than compressing the higher quality modality.
- We design controlled experiments on both multivariate and image datasets to directly mirror the theoretical comparison, and the results consistently validate the proposed principle in practical deep multimodal learning systems.

II. Theoretical Analysis of the Optimal Fusion Structure

In this section, we use concepts from statistical signal processing and information theory to theoretically analyze which decision fusion scheme performs better. To make the comparison more complete, we study the problem from two perspectives. We first analyze the fusion structure from the viewpoint of regression and signal estimation, where the quality of the fused observation is measured by Fisher information. We then analyze the same problem from the viewpoint of classification and signal detection, where the quality of the fused result is measured by the Bhattacharyya Distance. Both analyses lead to the same conclusion that under a fixed communication budget, assigning compression to the lower quality modality yields better final fusion performance than compressing the higher quality modality.



(a) Compressing modality L and fusing at S (b) Compressing modality S and fusing at L

Fig. 1. Two communication structures for bandwidth constrained two modality fusion

A. Perspective of regression and signal estimation

In signal estimation problems, we obtain information about the parameter from sample observations drawn from an underlying probability density function (pdf). In particular, we consider a random variable Z for which the pdf is $f(z|\theta)$, where θ is an unknown parameter and $\theta \in \Theta$, with Θ denoting the parameter space.

The Fisher information quantifies how much information the observation carries toward the estimation of θ, that is, a higher Fisher information leads to better estimation of the unknown parameter. The Fisher information for θ contained in the random variable Z is given by

$$I_Z(\theta) = -\int \big[\frac{\partial^2}{\partial \theta^2} \log f(z|\theta)\big] f(z|\theta) dz = -\mathbb{E}[l'']$$

which is equivalent to the reciprocal of the Cramer-Rao lower bound (CRLB).

Because Gaussian models are widely adopted in many signal processing settings, we consider the case where Z is observed through a Gaussian noisy channel

$$Z = \theta + v$$

where $v \sim \mathcal{N}(0, \sigma^2)$ is a noise term that follows a Gaussian distribution with mean 0 and variance σ^2. Hence, we have $f(z|\theta) \sim \mathcal{N}(\theta, \sigma^2)$. In this case, we have

$$l(z|\theta) = \log f(z|\theta) = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{(z-\theta)^2}{2\sigma^2}$$

and it follows that $l''(z|\theta) = -\frac{1}{\sigma^2}$ and $I_Z(\theta) = -E[l''(z|\theta)] = \frac{1}{\sigma^2}$.

We now use this result to compare the two communication structures in Fig. 1. Suppose modalities L and S observe the same underlying parameter through independent Gaussian noises. Let

$$L = \theta + v_L, \qquad S = \theta + v_S$$

where $v_L \sim \mathcal{N}(0, \sigma_L^2)$ and $v_S \sim \mathcal{N}(0, \sigma_S^2)$. We assume $\sigma_L^2 > \sigma_S^2$, which means that modality L is noisier and therefore has lower observation quality than modality S.

To model the effect of communication compression, we treat encoding as introducing an additional uncertainty term C to the compressed modality. Under the communication structure in Fig. 1(a), modality L is compressed and transmitted to modality S. The resulting total Fisher information is

$$I^a = \frac{1}{\sigma_L^2 + C} + \frac{1}{\sigma_S^2}.$$

Under the communication structure in Fig. 1(b), modality S is compressed and transmitted to modality L. The resulting total Fisher information is

$$I^b = \frac{1}{\sigma_L^2} + \frac{1}{\sigma_S^2 + C}.$$

Comparing these two expressions, we see that the same compression loss C has a smaller impact when it is applied to the noisier modality. Since $\sigma_L^2 > \sigma_S^2$, it follows that $I^a > I^b$. Therefore, from the perspective of estimation, compressing the lower quality modality leads to better final fusion performance under the same communication budget.

B. Perspective of classification and signal detection

We next analyze the same problem from the perspective of classification and signal detection. Our goal is again to compare which communication structure leads to better decision performance under the same compression budget. To evaluate classification quality, we consider a binary hypothesis testing problem [28] and use the Bhattacharyya Distance (BD) [29] as the performance metric. BD measures the separability between two distributions, and a larger BD indicates better classification performance.

The Bhattacharyya Distance is defined as

$$D_B(P, Q) = -\ln\Big(\sum_{x \in \mathcal{X}} \sqrt{P(x)Q(x)}\Big)$$

For the Gaussian case considered in this paper, let $L \sim \mathcal{N}(\mu_L, \sigma_L^2)$, and $S \sim \mathcal{N}(\mu_S, \sigma_S^2)$, where $\mathcal{N}(\mu, \sigma^2)$ denotes a normal distribution with mean μ and variance σ^2. Based on the above definition, the Bhattacharyya Distance between the two modalities can be written as

$$D_B(L, S) = \frac{1}{4}\frac{(\mu_L - \mu_S)^2}{\sigma_L^2 + \sigma_S^2} + \frac{1}{2}\ln\left(\frac{\sigma_L^2 + \sigma_S^2}{2\sigma_L\sigma_S}\right)$$

To incorporate the effect of communication compression, we use the same modeling idea as in the previous subsection and treat encoding as introducing additional uncertainty to the compressed modality. Under the communication structure in Fig. 1(a), modality L is compressed before transmission, so its effective variance increases to $\sigma_L^2 + C$, while modality S remains unchanged. As a result, the corresponding BD is given by

$$D_B^a(L, S) = \frac{1}{4}\frac{(\mu_L - \mu_S)^2}{\sigma_L^2 + \sigma_S^2 + C} + \frac{1}{2}\ln\left(\frac{\sigma_L^2 + \sigma_S^2 + C}{2\sigma_S\sqrt{\sigma_L + C}}\right)$$

Similarly, under the communication structure in Fig. 1(b), modality S is compressed before transmission, and the resulting BD becomes

$$D_B^b(L, S) = \frac{1}{4}\frac{(\mu_L - \mu_S)^2}{\sigma_L^2 + \sigma_S^2 + C} + \frac{1}{2}\ln\left(\frac{\sigma_L^2 + \sigma_S^2 + C}{2\sigma_L\sqrt{\sigma_S + C}}\right)$$

Comparing the above two expressions, we observe that the first term is identical in both cases. Therefore, the difference between the two communication structures is entirely determined by the logarithmic term. Under the assumption $\sigma_L^2 > \sigma_S^2$, it can be verified that

$$\sigma_S\sqrt{\sigma_L^2 + C} < \sigma_L\sqrt{\sigma_S^2 + C},$$

which implies that the logarithmic term in $D_B^a(L, S)$ is larger than that in $D_B^b(L, S)$. Consequently, we have $D_B^a(L, S) > D_B^b(L, S)$. This result shows that, from the perspective of classification, assigning compression to the modality with higher uncertainty leads to better final decision performance.

The estimation and classification analyses lead to the same conclusion that under a fixed communication budget, the better strategy is to compress the modality with higher uncertainty and preserve the modality with lower uncertainty. This theoretical result forms the basis for the experimental validation in the next section.

III. Experiment Validation

A. Experimental Setup

We evaluate our framework on two datasets: the RegDB dataset and a multivariate weather dataset [30,31]. The RegDB dataset consists of paired visible and thermal images, while the weather dataset is constructed from real world meteorological records with features partitioned into two complementary modalities. To ensure a fair comparison, we keep the total communication budget fixed and only vary which modality is assigned the compression bottleneck, thereby isolating the effect of modality wise compression.

For both datasets, we adopt a two modality deep fusion architecture. One modality serves as the fusion center (FC), while the other modality is compressed and transmitted under a fixed channel capacity. We alternatively assign compression to each modality while keeping the total channel width constant and compare the resulting prediction accuracy. To simulate heterogeneous modality quality, we inject additive Gaussian noise with different variances into each modality. A modality with larger noise variance is considered lower quality, while one with smaller variance is treated as higher quality. All models are trained using the Adam optimizer with a learning rate of 0.001. The RegDB models are trained for 100 epochs with a batch size of 64, while the weather dataset models are trained for 10 epochs with a batch size of 32.

These experiments are designed to evaluate whether the theoretical insight that compression applied to lower quality modalities incurs less performance degradation holds in practical deep multimodal fusion systems. Although the theoretical analysis is derived under simplified Gaussian assumptions, the experimental results demonstrate that the same principle generalizes to multimodal systems.

TABLE I
IMPACT OF COMPRESSED MODALITY ON WEATHER DATASET

Channel Width		Acc. (%)
Modality 1	Modality 2	
64(FC)	32(C)	85.54
32(C)	64(FC)	81.63
64(FC)	8(C)	85.69
8(C)	64(FC)	79.37
64(FC)	2(C)	84.79
2(C)	64(FC)	78.77
Baseline		
Modality 1		84.64
Modality 2		65.56

Modality 1 and Modality 2 denote two heterogeneous input modalities. FC denotes the fusion center, and C denotes the compressed signal transmitted to the other modality.
Baseline results are obtained by using a single modality with the same network configuration.

B. Multivariate Data Experiments

The two modalities are constructed by partitioning the feature space into two complementary subsets, each representing a distinct group of meteorological variables. This setup allows us to simulate heterogeneous sensing modalities with different predictive strengths. The single modality baselines are used to quantify the intrinsic quality of each modality, which serves as a reference for evaluating the impact of compression assignment.

We evaluate our proposed framework using a Multilayer Perceptron (MLP) encoder for the DC weather dataset. As a baseline, we train the same MLP architecture using single modality data only. As shown in Table I, the single modality baselines indicate a clear quality difference between the two modalities. Specifically, Modality 1 achieves a classification accuracy of 84.64%, whereas Modality 2 achieves only 65.56%. This shows that Modality 1 provides more reliable observations compared to Modality 2.

Under a fixed communication budget, a clear and consistent trend can be observed: assigning compression to the lower quality modality yields superior performance across all tested settings. This behavior aligns with the theoretical insight that compression induced information loss has a smaller impact on noisier modalities. Moreover, as the compression becomes more aggressive (i.e., the bottleneck width decreases), the performance gap between the two compression strategies becomes more pronounced. This suggests that high quality modalities are more sensitive to compression, making the preservation of their information increasingly critical under severe bandwidth constraints. For example, when the compressed channel width is 32, compressing the lower quality modality achieves 85.54% accuracy, compared to 81.63% when compressing the higher quality modality. As the width is reduced to 8, this gap increases to 6.32% (85.69% vs. 79.37%), and remains significant at width 2 (84.79% vs. 78.77%). These results consistently demonstrate the advantage of prioritizing high quality modalities for information preservation.

C. Image Data Experiments

We evaluate the proposed framework using a CNN based encoder on the RegDB dataset. As a baseline, we use the same CNN model with single modality inputs and apply additive Gaussian noise to the thermal and visible modality. We simulate heterogeneous modality quality by injecting additive Gaussian noise with two levels of variance: strong noise ($\mu = 0, \sigma = 1$) and weak noise ($\mu = 0.05, \sigma = 0.11$). This setup is intentionally designed to create a controlled quality asymmetry between the two modalities, so that we can directly compare the two compression assignments and examine whether the lower quality modality should consistently be the one compressed under a fixed communication budget.

As shown in Table II, the results on the RegDB dataset further reinforce the same principle under different noise configurations. Regardless of which modality is degraded, allocating more communication capacity to the higher-quality modality consistently yields better performance. When the thermal modality is heavily corrupted, the visible modality becomes the dominant information source. In this case, assigning compression to the thermal modality leads to significantly higher accuracy. For instance, at channel width 32, compressing the thermal modality achieves higher accuracy 91.99%, compared to 85.75% when compressing the visible modality. As compression becomes more aggressive, this the gap widens substantially, reaching over 30% at width 2. A symmetric trend is observed when the noise configuration is reversed. When the visible modality is degraded, the thermal modality becomes the dominant source of information. These observations confirm that the optimal compression strategy depends on modality quality rather than modality type, and that the proposed principle holds consistently across different settings.

IV. DISCUSSION

Our analysis, supported by both theoretical derivations and empirical results, shows that under a fixed communication bandwidth, compressing the lower quality modality leads to superior overall performance. The underlying reason lies in the asymmetric information contribution of heterogeneous modalities. When encoding is modeled as an additional uncertainty source, applying compression to the noisier modality induces

TABLE II
IMPACT OF COMPRESSED MODALITY ON REGDB DATASET.

Channel Width		Noise (μ, σ)		Acc.(%)
Thermal (T)	Visible (V)	Thermal (T)	Visible (V)	
64(FC)	32(C)	Strong	Weak	$85.75\% \pm 4.15$
32(C)	64(FC)	Strong	Weak	$91.99\% \pm 2.47$
64(FC)	8(C)	Strong	Weak	$78.00\% \pm 4.05$
8(C)	64(FC)	Strong	Weak	$94.13\% \pm 0.74$
64(FC)	2(C)	Strong	Weak	$67.65\% \pm 3.26$
2(C)	64(FC)	Strong	Weak	$99.39\% \pm 0.25$
64(FC)	32(C)	Weak	Strong	$52.27\% \pm 1.91$
32(C)	64(FC)	Weak	Strong	$48.93\% \pm 2.78$
64(FC)	8(C)	Weak	Strong	$67.83\% \pm 3.80$
8(C)	64(FC)	Weak	Strong	$48.21\% \pm 1.33$
64(FC)	2(C)	Weak	Strong	$80.37\% \pm 2.79$
2(C)	64(FC)	Weak	Strong	$40.53\% \pm 1.56$
Single-modality Baseline				
Thermal		Weak noise		$90.10\% \pm 2.07$
Visible		Weak noise		$98.64\% \pm 0.50$
Thermal		Strong noise		$36.02\% \pm 0.10$
Visible		Strong noise		$46.16\% \pm 1.55$

Strong: additive Gaussian noise with $(\mu, \sigma) = (0, 1)$.
Weak: additive Gaussian noise with $(\mu, \sigma) = (0.05, 0.11)$.
Baseline results are obtained by using a single modality with the same network configuration.

relatively smaller degradation in the total information. In contrast, compressing the more reliable modality reduces the effective information available for fusion. The experimental results further validate this principle. Across different datasets, noise configurations, and compression levels, allocating more transmission capacity to the higher quality modality consistently achieves superior accuracy. Moreover, the performance gap between correct and incorrect compression assignments widens as compression becomes more aggressive. This observation indicates that high quality modalities are more sensitive to compression induced information loss, especially under severe bandwidth constraints.

From a system design perspective, our findings provide a general principled guideline for communication aware multimodal architectures. Specifically, system designers should prioritize preserving the information of high quality modalities by allocating them sufficient communication bandwidth, while aggressively compressing lower quality modalities to improve overall fusion efficiency. This principle is particularly important in resource constrained environments such as edge intelligence and distributed sensing systems, where bandwidth allocation directly impacts the final decision performance. Moreover, our results suggest that adaptive bandwidth allocation strategies, which dynamically adjust compression levels based on modality reliability, can further enhance robustness and efficiency in practical deployments. Our current analysis is established under Gaussian assumptions and focuses on two modality settings. Extending the theoretical framework to non Gaussian models and multiple modalities remains an important direction for future research.

V. CONCLUSION

In this paper, we investigated modality compression strategies in bandwidth constrained multimodal fusion systems and provided both theoretical and empirical evidence for optimal compression design. Our analysis shows that, under Gaussian assumptions, compressing the modality with higher observation noise leads to improved overall fusion performance.

Extensive experiments on both multivariate and image datasets validate our theoretical findings, demonstrating that allocating more communication resources to higher quality modalities consistently improves decision accuracy. These results reveal a fundamental principle for multimodal system design: preserving reliable modalities is crucial for maintaining fusion performance under limited communication budgets.

The insights from this work provide a principled foundation for designing communication efficient multimodal architectures and have direct implications for real world applications such as edge intelligence, distributed sensing, and resource constrained AI systems.

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